# Kinematics

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, $t$ in seconds). Find expressions for its velocity $v$ and acceleration $a$.

$v = \frac{ds}{dt} = 3t^2 - 12t + 9$. $a = \frac{dv}{dt} = 6t - 12$.

**2** **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, $t$ in seconds). Find its velocity at $t = 2$ seconds.

$v = \frac{ds}{dt} = 6t^2 - 18t + 12$. At $t = 2$: $v = 6(4) - 18(2) + 12 = 24 - 36 + 12 = 0$ m/s.

**3** **[Foundational]** • **Target: The suvat equations for constant acceleration**

A car starts from rest and accelerates uniformly at $2.5$ m/s$^2$ for $8$ seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

$v = u + at = 0 + 2.5(8) = 20$ m/s. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.5)(64) = 80$ m.

**4** **[Foundational]** • **Target: The suvat equations for constant acceleration**

A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8$ m/s$^2$ (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.

$0 = 15^2 + 2(-9.8)s$, so $s = \frac{225}{19.6} \approx 11.48$ m.

**5** **[Foundational]** • **Target: The suvat equations for constant acceleration**

A cyclist decelerates uniformly from $12$ m/s to rest over a distance of $36$ meters. Find the deceleration, using $v^2 = u^2 + 2as$.

$v^2 = u^2 + 2as$: $0 = 12^2 + 2a(36) = 144 + 72a$, so $a = -2$ m/s$^2$ (a deceleration of $2$ m/s$^2$).

**6** **[Foundational]** • **Target: Velocity-time graphs**

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over $6$ seconds, then travels at constant velocity for a further $4$ seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.

Distance in first phase (trapezoid): $\frac{1}{2}(2 + 14)(6) = 48$ m. Distance in second phase (rectangle): $14 \times 4 = 56$ m. Total: $48 + 56 = 104$ m.

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**7** [Foundational] • Target: Velocity-time graphs

A particle decelerates uniformly from $v = 20$ m/s to rest over $5$ seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.

The graph forms a right triangle with base $5$ and height $20$. Distance $= \frac{1}{2}(5)(20) = 50$ m.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Motion under gravity

A stone is dropped from a height of $45$ meters. Using $g = 9.8$ m/s$^2$, find the time taken to reach the ground, and its speed on impact.

$s = \frac{1}{2}gt^2$: $45 = \frac{1}{2}(9.8)t^2$, so $t^2 = \frac{90}{9.8} \approx 9.18$, giving $t \approx 3.03$ s. $v = gt = 9.8(3.03) \approx 29.7$ m/s.

**9** [Intermediate] • Target: Motion under gravity

A ball is thrown vertically upward with initial speed $u = 24.5$ m/s. Using $g = 9.8$ m/s$^2$, find the total time taken to return to its starting point.

Time to reach the highest point: $t = \frac{u}{g} = \frac{24.5}{9.8} = 2.5$ s. By symmetry, the total time to return to the starting point is $2 \times 2.5 = 5$ s.

**10** [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 3t^2 - 2t$ (m/s). Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.

Displacement: $s = \int_0^3 (3t^2 - 2t)\,dt = [t^3 - t^2]_0^3 = 27 - 9 = 18$ m. Acceleration: $a = \frac{dv}{dt} = 6t - 2$; at $t = 3$: $a = 16$ m/s$^2$.

**11** [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 5t + 6$ (m/s). Find the times at which the particle is instantaneously at rest.

$t^2 - 5t + 6 = 0$, so $(t - 2)(t - 3) = 0$, giving $t = 2$ or $t = 3$ seconds.

**12** [Intermediate] • Target: Relative and combined motion

Two cars start 300 meters apart on a straight road and travel toward each other, one at $18$ m/s and the other at $12$ m/s, both at constant speed. Find the time until they meet.

Combined closing speed: $18 + 12 = 30$ m/s. Time $= \frac{300}{30} = 10$ seconds.

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**13** [Intermediate] • Target: Relative and combined motion

Two cars start at the same point on a straight road, one traveling at $25$ m/s. The second car starts $4$ seconds later, traveling at $30$ m/s. Find how long after the first car starts the second car catches up.

Let $t$ be the time (seconds) after the first car starts. First car's position: $25t$. Second car's position (starting 4 s later): $30(t-4)$. Setting equal: $25t = 30(t-4) = 30t - 120$, so $5t = 120$, giving $t = 24$ seconds.

**14** [Intermediate] • Target: A well-designed word problem: a train between two stations

This question has two parts. A train starts from rest at station A and accelerates uniformly at $0.5$ m/s$^2$ until it reaches a speed of $20$ m/s. It then travels at this constant speed before decelerating uniformly at $1$ m/s$^2$ to come to rest at station B. The total distance between the stations is $2400$ meters. (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.

(a) Acceleration phase: $t_1 = \frac{v-u}{a} = \frac{20}{0.5} = 40$ s. Distance: $s_1 = \frac{1}{2}(0+20)(40) = 400$ m. (b) Deceleration phase: $t_3 = \frac{20}{1} = 20$ s, distance $s_3 = \frac{1}{2}(20+0)(20) = 200$ m. Constant-speed distance: $2400 - 400 - 200 = 1800$ m, taking $t_2 = \frac{1800}{20} = 90$ s. Total journey time: $40 + 90 + 20 = 150$ seconds.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Relative and combined motion • Cross-Lesson: 8.01 Functions: Composite & Inverse

A train accelerates uniformly from rest to $30$ m/s in $20$ s, travels at constant speed $30$ m/s for $60$ s, and then decelerates uniformly to rest in $15$ s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

(a) $a_1 = \frac{v-u}{t} = \frac{30-0}{20} = 1.5$ m/s$^2$. (b) Total distance is the area of the trapezoid on the $v$-$t$ graph with parallel sides $60$ s and $20 + 60 + 15 = 95$ s, and height $30$ m/s: $s = \frac{60+95}{2} \times 30 = \frac{155}{2} \times 30 = 155 \times 15 = 2325$ m (c) Average speed.

**16** [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 9.01 Circular Measure

Two particles $P$ of mass $3$ kg and $Q$ of mass $5$ kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s$^2$: (a) find the acceleration $a$ of the system, (b) calculate the tension $T$ in the string, (c) find the force exerted by the string on the pulley.

(a) Write $F = ma$ for each particle in its direction of motion: For $Q$ (moving downwards): $5g - T = 5a$ For $P$ (moving upwards): $T - 3g = 3a$ Add the equations: $2g = 8a \Rightarrow a = \frac{2(9.8)}{8} = 2.45$ m/s$^2$ (b) Substitute $a$ into the equation for $P$: $T = 3g + 3a = 3(9.8 + 2.45) = 3(12.25) = 36.75$ N $\approx 36.8$ N (c) The pulley supports two vertical segments of string with tension $T$, so the total downward force on the pulley is.

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**17** [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A particle moves along a straight line such that its velocity in $\text{m/s}$ at time $t$ seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2$ s. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

(a) Acceleration: $a(t) = v'(t) = 6 - 4t$. At $t = 2$: $a(2) = 6 - 4(2) = -2$ m/s$^2$. (b) Net displacement is the integral of velocity: $\Delta s = \int_0^4 (6t - 2t^2)\,dt = \left[3t^2 - \frac{2}{3}t^3\right]_0^4 = (3(16) - \frac{2}{3}(64)) - 0 = 48 - \frac{128}{3} = \frac{16}{3} = 5\frac{1}{3}$ m. (c) Velocity vanishes when $6t - 2t^2 = 0 \Rightarrow 2t(3 - t) = 0 \Rightarrow t = 0$ and $t = 3$. For $0 \le t \le 3$, $v(t) \ge 0$.

**18** [Challenging] • Target: The suvat equations for constant acceleration • Cross-Lesson: 10.01 Series & Binomial Theorem

Practice 5.2: A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s for $0 \le t \le 4$. (a) Find the times at which the particle is momentarily at rest. (b) Calculate the net displacement of the particle over $[0, 4]$. (c) Calculate the total distance travelled by the particle over $[0, 4]$.

(a) $v(t) = 0 \Rightarrow 3(t^2 - 4t + 3) = 0 \Rightarrow 3(t - 1)(t - 3) = 0 \Rightarrow t = 1$ s and $t = 3$ s. (b) Displacement: $\Delta s = \int_0^4 (3t^2 - 12t + 9)\,dt = [t^3 - 6t^2 + 9t]_0^4 = (64 - 96 + 36) - 0 = 4$ m. (c) Total distance requires splitting the integral where velocity changes sign: Over $[0, 1]$: $v(t) > 0$; $\int_0^1 (3t^2 - 12t + 9)\,dt = [t^3 - 6t^2 + 9t]_0^1 = 4$ m. Over $[1, 3]$: $v(t)0$.

**19** [Challenging] • Target: Velocity-time graphs • Cross-Lesson: 12.01 Forces & Equilibrium

A block of mass $5$ kg is released from rest at the top of a rough plane inclined at $30^\circ$ to the horizontal. The length of the incline is $8$ m and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8$ m/s$^2$, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4$ m. Initial gravitational potential energy: $\text{PE} = mgh = 5(9.8)(4) = 196$ J 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}$ N. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}$ N. Work done against friction over distance $d = 8$ m: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896$ J 3. By.

**20** [Challenging] • Target: Motion under gravity • Cross-Lesson: 13.01 Newton's Laws

A stone is thrown vertically upwards with speed $15$ m/s from the top of a cliff $40$ m above sea level. Taking $g = 9.8$ m/s$^2$ and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

(a) At maximum height, $v = 0$. $u = +15$ m/s, $a = -9.8$ m/s$^2$: $v^2 = u^2 + 2as \Rightarrow 0 = 15^2 + 2(-9.8)s \Rightarrow 19.6s = 225 \Rightarrow s = \frac{225}{19.6} \approx 11.48$ m $\approx 11.5$ m (b) At the sea, the displacement relative to the launch point is $s = -40$ m. Use $s = ut + \frac{1}{2}at^2$: $-40 = 15t + \frac{1}{2}(-9.8)t^2 \Rightarrow -40 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 40 = 0$ Solve via quadratic formula.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Displacement, velocity, and acceleration

A particle moves so that its velocity at time $t$ s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

**a.** $t = 3$ s (excluding $t = 0$)

**b.** $t = 5$ s (excluding $t = 0$)

**c.** $t = 0$ s only

**d.** $t = 4$ s (excluding $t = 0$) ✓

Setting $v = 0$: $t^2 - 4t = 0 \Rightarrow t(t-4) = 0 \Rightarrow t = 4$ s (excluding $t = 0$)

**22** [Foundational] • Target: The suvat equations for constant acceleration

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s$^2$. Find its speed after travelling a displacement $s = 11$ m.

**a.** $\sqrt{79}$

**b.** $\sqrt{71}$

**c.** 75

**d.** $\sqrt{75}$ ✓

$v^2 = u^2 + 2as = (3)^2 + 2(3)(11) = 75 \Rightarrow v = \sqrt{75}$ m/s

**23** [Foundational] • Target: Velocity-time graphs

A particle has initial velocity $u = 4$ m/s and constant acceleration $a = 4$ m/s$^2$. Find its displacement after $t = 6$ s.

**a.** 24

**b.** 98

**c.** 94

**d.** **96** ✓

$s = ut + \frac{1}{2}at^2 = 4(6) + \frac{1}{2}(4)(6)^2 = 96$ m

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**24** [Foundational] • Target: Motion under gravity

A particle moves in a straight line such that its displacement $s$ metres from a fixed origin after $t$ seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of $t$ for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

**a.** t=1 s, s=6 m

**b.** t=2 s, s=4 m

**c.** **t=3 s, s=2 m ✓**

**d.** t=1 s, s=2 m

The displacement is given by $s = t^3 - 6t^2 + 9t + 2$. To find the velocity, we differentiate the displacement with respect to time: $v = \frac{ds}{dt} = \frac{d}{dt}(t^3 - 6t^2 + 9t + 2) = 3t^2 - 12t + 9$. The particle is instantaneously at rest when its velocity is zero, i.e., $v = 0$. So, we set the velocity equation to zero: $3t^2 - 12t + 9 = 0$. Divide by 3: $t^2 - 4t + 3 = 0$. Factorize the quadratic equation: $(t - 1)(t - 3) = 0$. This gives two possible values for $t$: $t = 1$ or $t = 3$. The question asks for 'the.

**25** [Foundational] • Target: Using calculus for variable acceleration

A particle of mass $2$ kg is acted upon by a constant force $\mathbf{F}$. At time $t = 0$, the particle is at the origin $(0, 0)$ and has velocity $\mathbf{v} = (3\mathbf{i} + 4\mathbf{j})$ m/s. At time $t = 2$ s, the particle's velocity is $\mathbf{v} = (7\mathbf{i} + 12\mathbf{j})$ m/s. Determine the magnitude of the constant force $\mathbf{F}$ acting on the particle.

**a.** $\frac{5\sqrt{17}}{2}$ N

**b.** $5\sqrt{17}$ N ✓

**c.** $\frac{5\sqrt{2}}{2}$ N

**d.** $5\sqrt{2}$ N

We are given the mass of the particle $m = 2$ kg. The initial velocity at $t = 0$ is $\mathbf{v}_0 = (2\mathbf{i} + 8\mathbf{j})$ m/s. The final velocity at $t = 2$ s is $\mathbf{v}_f = (7\mathbf{i} + 28\mathbf{j})$ m/s. The change in velocity is $\Delta\mathbf{v} = \mathbf{v}_f - \mathbf{v}_0$.

$\Delta\mathbf{v} = (7\mathbf{i} + 28\mathbf{j}) - (2\mathbf{i} + 8\mathbf{j}) = (7 - 2)\mathbf{i} + (28 - 8)\mathbf{j} = 5\mathbf{i} + 20\mathbf{j}$ m/s. The time interval i.

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**26** [Foundational] • Target: Finding when a particle is at rest or changes direction

A particle of mass 5 kg is at rest on a smooth horizontal surface. A constant horizontal force of magnitude 20 N is applied to the particle. What is the magnitude of the acceleration of the particle in $m/s^2$?

**a.** **4** ✓ **b.** 5

**c.** 20 **d.** 25

According to Newton's Second Law of Motion, the net force acting on an object is equal to the product of its mass and acceleration, i.e., $F = ma$. Given: Mass $m = 5$ kg, Force $F = 20$ N. We need to find the acceleration $a$. Rearranging the formula, $a = \frac{F}{m}$. Substituting the given values: $a = \frac{20\text{ N}}{5\text{ kg}} = 4\text{ m/s}^2$. Therefore, the magnitude of the acceleration of the particle is 4 $m/s^2$.

**27** [Foundational] • Target: Relative and combined motion

A particle has initial velocity $u = 10$ m/s and constant acceleration $a = 4$ $m/s^2$. Find its velocity after $t = 3$ s.

**a.** **22** ✓ **b.** 21

**c.** 23 **d.** 14

Applying Kinematics principles, $v = u + at = 10 + (4)(3) = 22$ m/s Therefore, the verified correct option is 22.

**28** [Foundational] • Target: A well-designed word problem: a train between two stations

A particle moves so that its velocity at time $t$ s is $v = 8t + 3$ m/s. Find the acceleration at $t = 0$ s.

**a.** 13 **b.** 10

**c.** **8** ✓ **d.** 6

$a = \frac{dv}{dt} = 8$. At $t = 0$: $a = 8$ $m/s^2$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Displacement, velocity, and acceleration

A particle moves so that its velocity at time $t$ s is $v = 2t$ m/s. Find an expression for its displacement $s$ in terms of $t$.

**a.** $3t^2 + C$

**b.** $t^2 + C$ ✓

**c.** 1+C

**d.** $2t^2 + C$

$s = \int v\,dt = t^2 + C$

**30** [Intermediate] • Target: The suvat equations for constant acceleration

A particle moves in a straight line such that its acceleration, $a$ m s$^{-2}$, at time $t$ seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point $O$ is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

**a.** $-37$ ✓

**b.** $-25$

**c.** $-13$

**d.** $-5$

1. **Find the velocity function $v(t)$:** $v(t) = \int a\,dt = \int (6t - 18)\,dt = 3t^2 - 18t + c$ Since the particle is at instantaneous rest when $t = 1$, $v(1) = 0$: $3(1)^2 - 18(1) + c = 0 \Rightarrow -15 + c = 0 \Rightarrow c = 15$ Thus, $v(t) = 3t^2 - 18t + 15$. 2. **Find when the particle is next at instantaneous rest:** Set $v(t) = 0$: $3(t^2 - 6t + 5) = 0 \Rightarrow 3(t - 1)(t - 5) = 0$ So the particle is at rest at $t = 1$ and next at $t = 5$. 3. **Find the displacement function $s(t)$:**.

**31** [Intermediate] • Target: Velocity-time graphs

A particle moves in a straight line such that its velocity, $v$ m s$^{-1}$, at time $t$ seconds is given by $v = 6t^2 - 18t + k$, where $k$ is a constant. Given that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first $2$ seconds.

**a.** 6 m ✓

**b.** 4 m

**c.** 5 m

**d.** 8 m

Step 1: Determine the unknown parameter $k$. The particle is momentarily at rest when $t = 2$, so $v(2) = 0$: $6(2)^2 - 18(2) + k = 0 \Rightarrow 24 - 36 + k = 0 \Rightarrow k = 12$ Step 2: Find all times when the particle is at rest to check for direction changes. $v(t) = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2) = 6(t - 1)(t - 2) = 0$ The particle is at rest at $t = 1$ and $t = 2$. In the interval $0 \le t0$; in the interval $1 < t < 2$, $v(t) < 0$. Step 3: Calculate the total distance traveled by integrating over each.

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**32** [Intermediate] • Target: Motion under gravity

A particle is in equilibrium under three collinear forces: two forces of $6$ N and $13$ N acting in one direction, and a third force $F$ acting in the opposite direction. Find $F$.

**a.** **19** ✓

**b.** 16

**c.** 21

**d.** 7

For equilibrium: $F = f_1 + f_2 = 6 + 13 = 19$ N

**33** [Intermediate] • Target: Using calculus for variable acceleration

At 08:00, a cargo vessel has position vector $(-7\mathbf{i} + 22\mathbf{j})$ km relative to a coastal radar station at the origin $O$, and moves with a constant velocity of $(15\mathbf{i} + 5\mathbf{j})$ km/h. At the same time (08:00), a high-speed patrol boat stationed at $(5\mathbf{i} + 6\mathbf{j})$ km sets off in a straight line at a constant speed $v$ km/h to intercept the cargo vessel. Find the minimum speed $v$ required for the patrol boat to intercept the cargo vessel, and the time at which interception occurs at this minimum speed.

**a.** **Minimum speed $15$ km/h, interception at 12:00** ✓

**b.** Minimum speed $15$ km/h, interception at 10:00

**c.** Minimum speed $5\sqrt{10}$ km/h, interception at 12:00

**d.** Minimum speed $12.5$ km/h, interception at 11:12

Let $t$ denote the time in hours elapsed after 08:00 (with $t > 0$). 1. **Position vectors at time $t$:** - Cargo vessel: $\mathbf{r}_C(t) = (-7\mathbf{i} + 22\mathbf{j}) + t(15\mathbf{i} + 5\mathbf{j}) = (15t - 7)\mathbf{i} + (5t + 22)\mathbf{j}$ - Patrol boat starts at $\mathbf{r}_P(0) = 5\mathbf{i} + 6\mathbf{j}$. 2. **Condition for interception:** For interception to occur at time $t$, the displacement of the patrol boat from its starting position must equal the relative position vector.

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**34** [Intermediate] • Target: Finding when a particle is at rest or changes direction

A supply launch $S$ departs from a base at the origin $O$ at time $t = 0$ to intercept an automated cargo barge $C$. At $t = 0$, the barge has position vector $\mathbf{r}_0 = (16; 12)$ km relative to $O$ and maintains a constant velocity of $\mathbf{v}_C = (-15; 0)$ km/h. The launch travels in a straight line with constant velocity $\mathbf{v}_S$ and constant speed $v$ km/h. Determine the minimum speed $v$ required for the launch to successfully intercept the barge.

**a.** 9 km/h ✓

**b.** 12 km/h

**c.** 15 km/h

**d.** 7.2 km/h

To intercept the cargo barge at some time $t > 0$, the position vectors must satisfy: $\mathbf{r}_S(t) = \mathbf{r}_C(t) \Rightarrow t\mathbf{v}_S = \mathbf{r}_0 + t\mathbf{v}_C$ Dividing by $t$ gives: $\mathbf{v}_S = \mathbf{v}_C + \frac{1}{t}\mathbf{r}_0 = (-15; 0) + \frac{1}{t}(16; 12) = \left(-15 + \frac{16}{t}; [4pt]\frac{12}{t}\right)$ Let $u = \frac{1}{t} > 0$. The square of the speed of the launch is.

**35** [Intermediate] • Target: Relative and combined motion

Two particles, A and B, move in the xy-plane. Their position vectors at time $t$ seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

**a.** $\sqrt{26}$ ✓

**b.** $2\sqrt{13}$

**c.** $3\sqrt{10}$

**d.** $4\sqrt{5}$

The position vectors of particle A and particle B at time $t$ are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time $t$ is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time $t$ is the magnitude of this displacement vector.

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**36** [Intermediate] • Target: A well-designed word problem: a train between two stations

A particle moves in a straight line such that its acceleration, $a$ m s$^{-2}$, at time $t$ seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is $5$ m s$^{-1}$. Find the velocity of the particle when $t = 3$.

**a.** $15$ m s$^{-1}$

**b.** $19$ m s$^{-1}$

**c.** $21$ m s$^{-1}$ ✓

**d.** $27$ m s$^{-1}$

To find the velocity $v$, integrate the acceleration function with respect to $t$: $v = \int a\,dt = \int (6t - 4)\,dt = 3t^2 - 4t + C$ Using the initial condition that $v = 5$ when $t = 1$: $3(1)^2 - 4(1) + C = 5 \Rightarrow -1 + C = 5 \Rightarrow C = 6$ Thus, the velocity function is: $v(t) = 3t^2 - 4t + 6$ Substitute $t = 3$: $v(3) = 3(3)^2 - 4(3) + 6 = 27 - 12 + 6 = 21$ m s$^{-1}$ Common misconceptions: - Omitting the constant of integration $C$ gives.

**37** [Intermediate] • Target: Displacement, velocity, and acceleration

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude $20$ N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is $120^\circ$, find the magnitude of the third force.

**a.** 18

**b.** **20** ✓

**c.** 25

**d.** 22

By symmetry (each pair at $120^\circ$), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 20$ N (standard $120^\circ$-$120^\circ$-$120^\circ$ symmetric equilibrium).

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: The suvat equations for constant acceleration • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle of mass $6$ kg moving at $6$ m/s collides and sticks to a particle of mass $6$ kg moving at $0$ m/s (same line of motion). Find the common velocity after collision.

**a.** 2

**b.** 4

**c.** 6

**d.** **3** ✓

By conservation of momentum: $m_1u_1 + m_2u_2 = (m_1 + m_2)v \Rightarrow (6)(6) + (6)(0) = (12)v \Rightarrow v = 3$ m/s

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**39** [Challenging] • Target: Velocity-time graphs • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass $2$ kg and $5$ kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s$^2$, find the acceleration of the system.

**a.** **4.20** ✓

**b.** 8.40

**c.** 4.70

**d.** 3.70

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(5-2)(9.8)}{2+5} = 4.20$ m/s$^2$

**40** [Challenging] • Target: Motion under gravity • Cross-Lesson: 13.01 Newton's Laws

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s in $t = 6$ s. Find its acceleration.

**a.** 1

**b.** 3

**c.** **2** ✓

**d.** 12/7

$a = \frac{v - u}{t} = \frac{14 - (2)}{6} = 2$ m/s$^2$

**41** [Challenging] • Target: Using calculus for variable acceleration • Cross-Lesson: 14.01 Energy, Work & Power

Relative to a fixed origin $O$, ship $A$ has an initial position vector of $(24\mathbf{i} + 10\mathbf{j})$ km and moves with a constant velocity of $(-2\mathbf{i} + 5\mathbf{j})$ km h$^{-1}$. Ship $B$ has an initial position vector of $(-20\mathbf{i} - 18\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 12\mathbf{j})$ km h$^{-1}$, where $u$ is a constant. Given that ship $B$ intercepts ship $A$, find the speed of ship $B$.

**a.** 15 km h$^{-1}$ ✓

**b.** 9 km h$^{-1}$

**c.** 21 km h$^{-1}$

**d.** 34 km h$^{-1}$

Let $t$ be the time in hours after the ships begin moving. The position vector of ship $A$ at time $t$ is: $\mathbf{r}_A(t) = (24\mathbf{i} + 10\mathbf{j}) + t(-2\mathbf{i} + 5\mathbf{j}) = (24 - 2t)\mathbf{i} + (10 + 5t)\mathbf{j}$ The position vector of ship $B$ at time $t$ is: $\mathbf{r}_B(t) = (-20\mathbf{i} - 18\mathbf{j}) + t(u\mathbf{i} + 12\mathbf{j}) = (-20 + ut)\mathbf{i} + (-18 + 12t)\mathbf{j}$ For ship $B$ to intercept ship $A$, their positions must be identical at some time $t > 0$, so.

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**42** [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 15.01 Representation of Data

At 14:00, a coastguard radar station detects two vessels, $A$ and $B$. Relative to the radar station, vessel $A$ has position vector $(-4\mathbf{i} + 6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i} + 5\mathbf{j})$ km h$^{-1}$. Vessel $B$ has position vector $(20\mathbf{i} - 4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 10\mathbf{j})$ km h$^{-1}$, where $u$ is a constant. Given that vessel $B$ intercepts vessel $A$, find the value of $u$.

**a.** $u = -4$ ✓

**b.** $u = 4$

**c.** $u = -2$

**d.** $u = -8$

Let $t$ be the time in hours after 14:00 when the interception occurs. The position vector of vessel $A$ at time $t$ is: $\mathbf{r}_A(t) = (-4 + 8t)\mathbf{i} + (6 + 5t)\mathbf{j}$ The position vector of vessel $B$ at time $t$ is: $\mathbf{r}_B(t) = (20 + ut)\mathbf{i} + (-4 + 10t)\mathbf{j}$ For interception to occur, $\mathbf{r}_A(t) = \mathbf{r}_B(t)$. Equating the $\mathbf{j}$-components: $6 + 5t = -4 + 10t \Rightarrow 5t = 10 \Rightarrow t = 2$ hours Equating the $\mathbf{i}$-components and substitu.

**43** [Challenging] • Target: Relative and combined motion • Cross-Lesson: 16.01 Permutations & Combinations

A particle moves in a straight line with acceleration $a(t) = 6t - 8$ m/s$^2$. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?

**a.** $t = 4$ s

**b.** $t = \frac{4}{3}$ s

**c.** $t = \frac{8}{3}$ s ✓

**d.** $t = 8$ s

Velocity $v(t) = \int (6t - 8)\,dt = 3t^2 - 8t + C$. Since $v(0) = 0 \Rightarrow C = 0$. Momentarily at rest when $v(t) = 0 \Rightarrow t(3t - 8) = 0 \Rightarrow t = \frac{8}{3}$ s.

**44** [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 17.01 Discrete Random Variables

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

**a.** **60 J** ✓

**b.** 19 J

**c.** 11 J

**d.** 3.75 J

The work done by a constant force is given by the formula: Work Done = Force × Distance × cos(θ) where θ is the angle between the force and the direction of motion. In this case, the force and the distance are in the same direction, so θ = 0°. Therefore, cos(0°) = 1. Work Done = 15 N × 4 m × 1 Work Done = 60 J

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**45** [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 18.01 Binomial Distribution

Three concurrent forces keep a particle in equilibrium, each pair separated by $120^\circ$, and two of the forces have magnitude $28$ N. Find the magnitude of the third force.

**a.** 26

**b.** 30

**c.** 33

**d.** **28** ✓

With all three angles equal to $120^\circ$, Lami's theorem gives $\frac{F_1}{\sin 120^\circ} = \frac{F_2}{\sin 120^\circ} = \frac{F_3}{\sin 120^\circ} \Rightarrow F_3 = F_1 = 28$ N.

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